game46

after a long time, the 20 microfarad capacitor acts as an open circuit

Section	Capacitors	Inductors
Differential i-v	$i = C\frac{dv}{dt}$	$V = L\frac{di}{dt}$
Integral i-v	$v_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(\tau)d\tau$	$i_C(t) = \frac{1}{L}\int_{-\infty}^{t} v_L(\tau)d\tau$
DC Equivalent	Open Circuit	Short Circuit
Two in series	$C_{eq} = \frac{C_1 C_2}{C_1+C_2}$	$L_{eq} = L_1 + L_2$
Two in parallel	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1 L_2}{L_1+L_2}$
Stored energy	$W_C = \frac{1}{2}Cv_C^2$	$W_L = \frac{1}{2}Li_L^2$

When the switch flips, the 20 microfarad capacitor acts as a 300V voltage source sending current up and into the righthand side, into the 5 henry inductor and 800 ohm equivalent resistance resistor network

$$V_C(0^-) = 300V$$
$$i_C(0^-) = 0$$
Current and current cannot change instantaneously
$$i_C(0^+) = 0A$$
$$i_L(0^-) = 0A$$
$$V_C(0^+) = 300V$$
$$R_{eq} = 800\Omega$$
$$\alpha = \frac{R}{2L} = 80$$
$$\omega_0 = \sqrt{\frac{1}{0.00001}} = 100$$
$$\alpha < \omega_0 \therefore i(t) = B_1 e^{-\alpha t}\cos\omega_d t + B_2 e^{-\alpha t}\sin\omega_d t, \;\; t \geq 0$$
$$i(0^+) = B_1 = I_0 = 0A$$
$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 60$$
$$i(t) = B_2 e^{-80t}\sin 60t$$
$$-\alpha B_1 + \omega_d B_2 = \frac{1}{L}(-RI_0 - V_0)$$
$$60B_2 = \frac{1}{5}(-0 - (-300))$$
$$60B_2 = 60$$
$$B_2 = 1$$
$$i(t) = e^{-80t}\sin 60t$$
$$V = L\frac{di}{dt}$$
$$= 300e^{-80t}\cos(60t) - 400e^{-80t}\sin(60t)$$
$$v_a(t) = v_{600}(t) + v_L(t) = 600i(t) + 300e^{-80t}\cos(60t) - 400e^{-80t}\sin(60t)$$
$$= \boxed{300e^{-80t}\cos(60t) + 200e^{-80t}\sin(60t)V}$$